DFENet for Domain Adaptation-Based Remote Sensing Scene Classification

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***Abstract*— Domain adaptation scene classification refers to the task of scene classification where the training set (called source domain) has different distributions from the test set (called target domain). Although remarkable results have been reported, the misalignment of source and target domain features still remains a big challenge when the large intraclass variances of remote sensing images encounter the insufficient exploration of discriminative feature representations for both domains. To address this challenge, a novel domain feature enhancement network (DFENet) is proposed to adaptively enhance the discriminative ability of the learned features for dealing with the domain variances of scene classification. Specifically, an adaptive context-aware feature refinement (CAFR) module is first designed to automatically recalibrate global and local features by explicitly modeling interdependencies between the channel and spatial for each domain. Then, a multilevel adversarial dropout (MAD) module is further designed to strengthen the generalization capability of our network by adaptively reconfiguring the sparsity of the feature level and decision level in the target domain. The cooperation of CAFR module and MAD module formulates a unique DFENet that can be learned in an end-to-end manner. Comprehensive experiments show that our proposed method is better than state-of-the-art methods on Merced → RSSCN7, AID → RSSCN7, NWPU → RSSCN7, RSSCN7 → Merced, RSSCN7 → AID, and RSSCN7 → NWPU datasets.**

***Index Terms*— Domain adaptation scene classification, domain feature enhancement network (DFENet), remote sensing images (RSIs).**

I. Introduction

With the rapid development of remote sensing technology, the ability to acquire remote sensing images (RSIs) is growing exponentially, which greatly promotes understanding the contents of RSIs [1]–[4]. Scene classification is a basic and challenging task for remote sensing analysis interpretation, which plays a crucial role in the automatic processing of remote sensing information [5]–[8].

Scene classification aims to classify scene images into different semantic categories [9]–[11]. As pointed out in [8], most scene classification methods are trained in a fully supervised manner, which often requires a great quantity of data with annotation information. Notwithstanding the rapid growth of the quantity and quality of RSIs, manually labeled annotations for abundant datasets need a tremendous amount of time and labor. To remedy this, an appealing feasible method is to use the ready-made labeled remote sensing data to transfer unlabeled data [12]–[15], which makes great efforts to use the knowledge of labeled source domain to help for learning in a related but unlabeled target domain. Compared with the source domain, the remote sensing data of the target domain may be collected with different geographic locations, sensors, imaging conditions, and so on. As a result, there is usually a large domain discrepancy between the source and target domains. In this article, we focus on the scene classification problem under the domain adaptation setting. Domain adaptation does not need annotation information in the target domain, which can significantly reduce labor costs.

Currently, a common domain adaptation method is the maximum mean discrepancy and its higher-order kernel variants [16]–[18]. These works mainly aim to minimize the divergence between the source and target domains. The prevalent adversarial learning method [19], [20] for domain adaptation through the game of the generator and discriminator can maximize the confusion between the source and target representations, thus domain invariant feature representations are learned. Among them, the work [17] designed a central moment discrepancy, which can minimize the domain discrepancy of latent feature representations. The work [19] proposed an adversarial discriminative domain adaptation framework. The framework first used the label information of the source domain to learn a priori knowledge, and then the target data was mapped to separate encoding in the same space by using the asymmetric mapping of domain adversarial loss.

Although significant performance improvements have been achieved by the above methods, the misalignment of source and target domain feature distributions is still a big challenge when the large intra-class variances encounter the insufficient exploration of discriminative feature representations for both domains. What is worse, RSIs usually exhibit complex spatial structures with intraclass variances. For instance, in the first column of the source and target domain images in Fig. 1, the scene category of these images is "Dense residential"; however, the scenes in these images are very different in spatial layout. Other columns of the source and target domain

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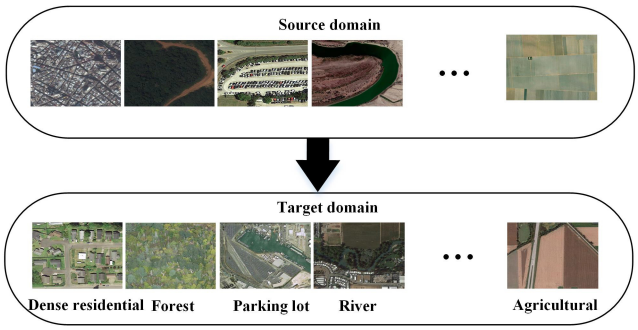


Fig. 1. RSIs scene classification with domain adaptation from source domain to target domain.

images in Fig. 1 have a similar scene form. The large intraclass variances directly lead to the large domain gap between the source and target domains. Obviously, it is very difficult to learn a scene classification model with strong discriminative ability from RSIs.

To overcome the aforementioned limitations of domain adaptation scene classification, we innovatively design an adaptive context-aware feature refinement (CAFR) module to capture the context dependencies from both global and local perspectives for each domain, which contains a dynamic global context-aware learning module (DG-CALM) and a dynamic local context-aware learning module (DL-CALM). Specifically, the DG-CALM is designed for global feature discriminative learning through a channel-wise self-gating mechanism to "adaptively" recalibrate the generated features and suppress less useful ones. The DL-CALM is developed for the optimization of local information by a dynamic spatial-wise self-gating mechanism to "automatically" emphasize the valuable feature information. The cooperation of DG-CALM and DL-CALM forms a unique end-to-end CAFR module to dynamically activate global and local feature responses, and separately performs feature discriminative learning for each domain. As a consequence, our method offers a promise, that is, domain discriminative learning features would be preserved while domain features can be sufficiently learned.

To further strengthen the domain feature learning in RSIs, we develop a multilevel adversarial dropout (MAD) module to improve the generalization capability of our network by automatically reconfiguring the sparsity of the feature level and decision level. This operation separately drops low-quality discriminative features by the generated adversarial dropout mask in feature level and decision level, which remarkably heightens the discriminative learning of features.

The collaboration of CAFR and MAD constructs a unique end-to-end domain feature enhancement network (DFENet), as shown in Fig. 2. The main contributions of this article can be summarized as follows.

1) By studying the large intraclass variances of RSIs, a novel yet effective CAFR module is proposed to adaptively extract global and local discriminative learning features for the source and target domains.
2) We introduce an MAD module in the target domain, which can effectively strengthen discriminative learning of features to cut down the domain discrepancy.
3) Experimental results on six adaptation scenarios show that the proposed method achieves compelling results in the scene classification task.

The remainder of this article is organized as follows. Some related works are briefly reviewed in Section II. Section III describes our proposed method in detail. In Section IV, we provide thorough experimental results and analyses. Finally, Section V provides the conclusion.

II. Related Works

A. Domain Adaptation Scene Classification

Recently, domain adaptation has achieved a series of successes in natural images [21]–[24], which has received more and more attention. Although the application of domain adaptation in natural image classification has achieved exciting results, these methods cannot be directly applied to RSIs. Compared with natural images in domain adaptation settings, scene classification from RSIs is particularly challenging because RSIs usually show large intra-class variances and interclass similarity [25]. In addition, the difference of RSIs in size and rotation angle further aggravates the problem [26]. In practice, based on domain adaptation remote sensing scene classification is concerned and studied more commonly to reduce domain shift. The work [5] proposed to apply the pretrained convolutional neural network to separately produce a new initialization feature for the source and target images, which forward into an extra network for further learning to mitigate the domain discrepancy in RSIs scene classification. The work [12] proposed an asymmetric adaptation neural network for domain adaptation scene classification. In [6], a method based on the generative adversarial network is constructed for domain adaptation semi-supervised scene classification in RSIs. The work [27] embedded a subspace alignment layer into the CNN model to settle the domain discrepancy problem in RSIs. The work [15] proposed a correlation subspace dynamic distribution alignment approach for domain adaptation scene classification. The work [28] proposed a multisource compensation network to deal with distribution discrepancy and category incompleteness. Although the above methods have achieved satisfactory results, they fail to explicitly consider feature discriminative learning, thus the source and target feature distributions may not be aligned.

B. Context-Aware Embedding

Context information has been successfully developed in perceptual tasks such as image classification [29], [30]. It plays an important role in enhancing feature representations. Woo *et al.* [31] constructed a convolutional block attention module to increase feature representation power by effectively capturing the global and local semantic context information. Hu *et al.* [32] formulated the "context" modules as the composition of the gather and excite operator, where the gather operator aggregates the contextual information across large neighborhoods of each feature map and the excited operator modulates the feature maps by adjusting the aggregation. Bello *et al.* [33] employed an augment convolutional operator with this self-attention mechanism to capture long-range

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contextual information. However, the aforementioned methods need a great deal of training data with annotation information, and data labeling is a labor and money-consuming task. For this purpose, some recent work about the domain adaptation paradigm has been proposed to capture contextual information. Yang *et al.* [34] designed a cross-domain attention network to adapt both spatial and channel context dependencies between the source and target domains. Kang *et al.* [35] proposed an attention alignment method on the target convolutional layer to reveal the knowledge shared by the source domain. Li *et al.* [36] designed a new spatial attention pyramid network to fully exploit context information around the objects for domain adaptation. Besides, Li *et al.* [37] proposed a generalized domain conditioned adaptation network to transfer domain low-level and domain high-level features simultaneously for each domain.

Generally speaking, there are few concerns about how to take full advantage of the context-aware to enhance feature representation under domain adaptation. In addition, existing domain adaptation methods simply combine global context features with local context features to perform distribution alignment. Compared with them, we train the scene classification model under domain adaptation and make the best of channel-wise and spatial-wise context-aware in all locations to dynamically recalibrate the global and local feature responses by a self-gating mechanism, which enhances the mining of high-quality overall objects.

III. Methodology

A. Context-Aware Feature Refinement Module

We design a CAFR module for the RSIs scene classification task. Our goal is to model discriminative learning features by adaptively recalibrating global and local feature responses for the source and target domains.

1) Dynamic Global Context-Aware Learning Module: In this installment, our goal is to capitalize on a self-gating mechanism to dynamically recalibrate the global features for each domain, such that the regions with discriminative context information features can be discovered. The architecture of DG-CALM is shown in Fig. 3(a). Given $A \in \mathbb{R}^{C\times H\times W}$ (C is the number of channels, and $H \times W$ denotes the image size), both average-pooling and max-pooling operations are applied to generate average-pooled features and max-pooled features, respectively. These are predicted by the followed two share convolutional layers, and producing two feature vectors $P_a \in \mathbb{R}^{C\times 1\times 1}$ and $P_m \in \mathbb{R}^{C\times 1\times 1}$. We combine the generated feature vectors P_a and P_m by using element-wise summation to produce a channel attention mask vector $M \in \mathbb{R}^{C\times 1\times 1}$. M is then fed into a convolution layer to obtain vector $N \in \mathbb{R}^{2C\times 1\times 1}$, we reshape N to $2 \times C \times 1 \times 1$, N is normalized by using the softmax function to compute the channel-wise context-aware self-gating $\mathcal{Y}_{\mathbf{c}} \in \mathrm{R}^{2\times C\times 1\times 1}$

$$\mathcal{Y}_{\mathbf{c}} = \frac{\exp(p_i)}{\sum_i^{N_t}\exp(p_i)} \tag{1}$$

where $p_i \in N$. We reshape $\mathcal{Y}_{\mathbf{c}}$ to $2C \times 1 \times 1$, $V_c = [P_a, P_m]$, the dynamic global context-aware feature F_c is obtained by element-wise multiplication

$$F_c = \mathcal{Y}_{\mathbf{c}} \otimes V_c. \tag{2}$$

The feature F_c is merged using element-wise summation to produce our dynamic global context-aware mask. Then, using a sigmoid operation to generate the normalization feature $M_c \in \mathbb{R}^{C\times 1\times 1}$. The final fusion feature $W_c \in \mathbb{R}^{C\times H\times W}$ is obtained by element-wise multiplication

$$W_c = M_c \otimes A. \tag{3}$$

2) Dynamic Local Context-Aware Learning Module: In this section, we aim to exploit a dynamic self-gating mechanism to recalibrate local features for the source and target domains. Thus, the local discriminative context region features can be discovered to the greatest extent.

The architecture of DL-CALM is shown in Fig. 3(b). Given $B \in \mathbb{R}^{C\times H\times W}$, both average-pooling and max-pooling operations are applied to generate average-pooled features $Q \in \mathbb{R}^{1\times H\times W}$ and max-pooled features $K \in \mathbb{R}^{1\times H\times W}$, respectively. These are predicted by the following two convolutional layers to produce a local context-aware mask $P \in \mathrm{R}^{2\times H\times W}$. We reshape P to $2\times 1\times H\times W$, P is normalized by using the softmax function to compute the spatial-wise context-aware self-gating $\mathcal{Y}_s \in \mathbb{R}^{2\times 1\times H\times W}$

$$\mathcal{Y}_s = \frac{\exp(p_i)}{\sum_i^{N_s}\exp(p_i)} \tag{4}$$

where $p_i \in P$. We reshape $\mathcal{Y}_s$ to $2 \times H \times W$, $V_s = [Q, K]$, the dynamic local context-aware feature F_s is obtained by element-wise multiplication

$$F_s = \mathcal{Y}_s \otimes V_s. \tag{5}$$

The feature F_s is concatenated and convolved to produce a 2-D dynamic local context-aware feature 2-D dynamic local context-aware feature. Then, the sigmoid operation is used to produce the normalization feature $M_s \in \mathbb{R}^{1\times H\times W}$. The final fusion feature $W_s \in \mathbb{R}^{C\times H\times W}$ is obtained by element-wise multiplication

$$W_s = M_s \otimes B. \tag{6}$$

3) Aggregation of DG-CALM and DL-CALM: To fully exploit channel-wise and spatial-wise context semantic information, we aggregate DG-CALM and DL-CALM to formulate a unique end-to-end CAFR module. Given a feature map $A \in \mathbb{R}^{C\times H\times W}$ as input, DG-CALM sequentially generates a 2-D channel-wise dynamic context-aware feature map $W_c \in \mathbb{R}^{C\times H\times W}$ and a 2-D spatial-wise dynamic context-aware feature map $W_s \in \mathbb{R}^{C\times H\times W}$, as shown in Fig. 2(c). The total dynamic context-aware process can be summarized as

$$T = W_s(W_c) \tag{7}$$

where T is the output of the final refined discriminative features.

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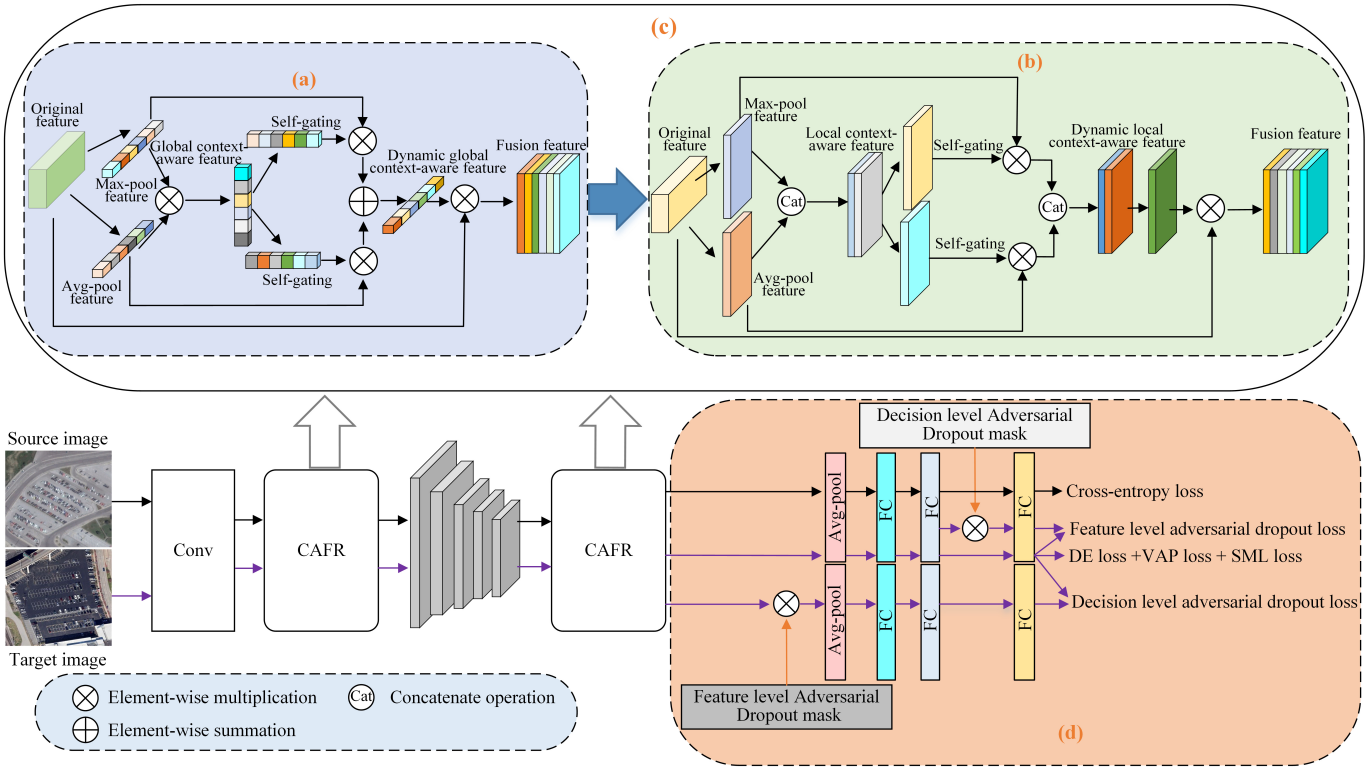


Fig. 2. Overview of the proposed framework. Purple arrows are used for target domain, and black arrows are used for source domain. (a) DG-CALM. (b) DL-CALM. (c) CAFR. (d) MAD.

B. Multilevel Adversarial Dropout Module

In this section, we develop an MAD module for domain adaptation scene classification, which automatically reconfigures the neural network by increasing the sparsity of the neural network as illustrated in Fig. 2(d). More specifically, the MAD module consists of a supervised adversarial dropout and an unsupervised adversarial dropout. Since our target domain has no ground truth labels, here we only leverage unsupervised adversarial dropout to maximize the divergence between two independent predictions to an input. For target domain data, unsupervised adversarial dropout takes a data point and generates two divergent predictions. In fact, our goal is to reduce the potential ambiguous predictions by minimizing the large intra-class variances of RSIs. For this reason, we introduce a decision level unsupervised adversarial dropout (DUAD) and a feature level unsupervised adversarial dropout (FUAD).

We first define a neural network G with a dropout layer. More specifically, we utilize the dropout layer to divide G into a feature extractor G_1 and a classifier G_2

$$G(x;\beta) = G_2(\beta \odot G_1(x)) \tag{8}$$

where $\odot$ is an element-wise multiplication, and β is a dropout mask.

1) Decision-Level Unsupervised Adversarial Dropout: Due to the large intraclass variances of RSIs, we apply unsupervised adversarial dropout to reconfigure the decision level of the network to generate a decision level adversarial dropout mask, which strengthens the discriminative learning of prediction features. According to the random dropout mask β^t, we define a DUAD mask β^{duad}

$$\begin{aligned}\beta^{\text{duad}} &= \underset{||\beta-\beta^t||_2\le\delta H}{\arg\max}\ E\big[G(x;\beta^t), G(x;\beta)\big] \\ &= \underset{||\beta-\beta^t||_2\le\delta H}{\arg\max}\ E\big[G_2(\beta^t \odot G_1(x)), G_2(\beta \odot G_1(x))\big]\end{aligned} \tag{9}$$

where $E[\cdot,\cdot]$ is a divergence function, δ is a hyperparameter that modulates the strength of the adversarial perturbation, and H is the dropout layer dimension. At each training process, we identify the optimal DUAD mask β^{duad} for the current model $G_2(\beta^t \odot G_1(x))$ and train the model by minimizing the regularization term to make it robust to this dropout perturbation.

To identify β^{duad}, we approximate the divergence function $E[G_2(\beta^t \odot G_1(x)), G_2(\beta \odot G_1(x))]$ with the first order of the Taylor expansion by relaxing the domain space of β^{duad}. After the approximation, we find a DUAD mask β^{duad} by solving an integer-programming problem.

In order to optimize β^{duad}, we maximize the following divergence function under the constraint $||\beta^t - \beta^{\text{duad}}||_2 \le \delta H$

$$E(x,\beta;\beta^t) = E\big[G_2(\beta^t \odot G_1(x)), G_2(\beta \odot G_1(x))\big]. \tag{10}$$

We utilize the first order of the Taylor expansion to approximate the above divergence function, which is a common step in the integer programming research

$$E(x,\beta;\beta^t) \approx E(x,\beta^0;\beta^t) + (\beta-\beta^0)J(x,\beta^0) \tag{11}$$

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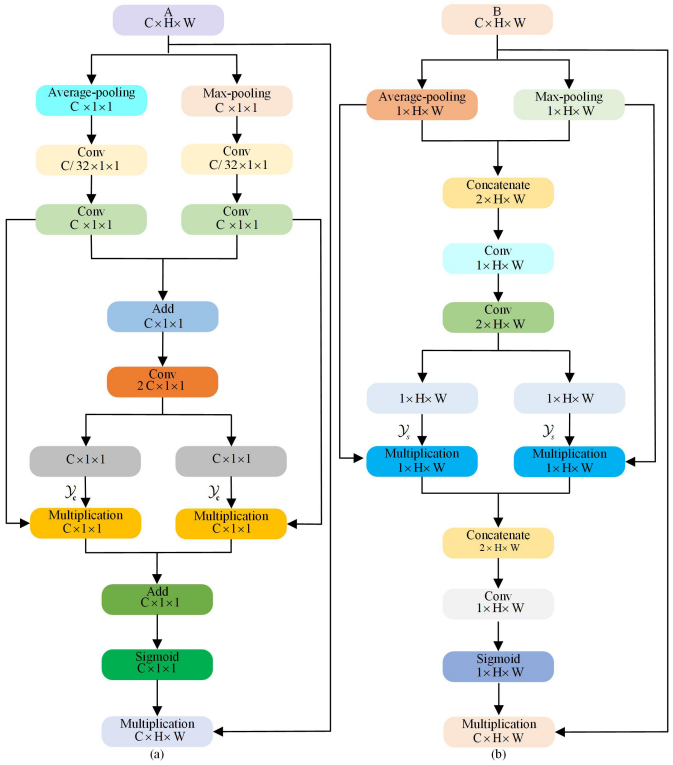


Fig. 3. Illustration of adaptive CAFR module architecture. (a) DG-CALM aims to dynamically capture global feature dependencies to recalibrate global features. (b) DL-CALM aims to automatically capture the local context of a visual scene to recalibrate local features.

where $J(x, \beta^0)$ is the Jacobian vector

$$J(x, \beta^0) := \nabla_\beta E(x, \beta; \beta^t)|_{\beta=\beta^0}. \tag{12}$$

The above formula linearly optimizes the divergence function by controlling β. Therefore, according to the Taylor expansion with respect to β, we express the divergence function as follows:

$$E(x, \beta; \beta^t) \propto \sum_i \beta_i J_i(x, \beta^0) \tag{13}$$

where $J_i(x, \beta^0)$ is the ith element of $J(x, \beta^0)$. Since $J_i(x, \beta^0)$ cannot continue to be derived, we introduce a new $J(x, \beta^0)$ that further improves the adversarial dropout by $\odot$ and h

$$J(x, \beta^0) \approx h \odot \nabla_h E(x, \beta^0; \beta^t) \tag{14}$$

where h is the output vector of the subnetwork $G_1(x)$. After calculating the $J(x, \beta^0)$, we use a greedy algorithm to find β^{duad}, which estimates $J(x, \beta^0)$ once and iteratively flips a dropout element shown to be highly effective to the distance term in view of the $J(x, \beta^0)$. The adversarial dropout β^{duad} with the greedy algorithm is shown in the following equation:

$$\beta^{\text{duad}} = \text{Flip}_{||\beta^t - \beta^0|| \le \delta H}(\beta^t, J(x, \beta^0)) \tag{15}$$

where we apply a flip function, which can change the dropout mask β^t in the descending order of $(1 - 2\beta^t) \odot J(x, \beta^0)$ until the constraint $||\beta^t - \beta^0||_2 \le \delta H$ is converged.

2) Feature-Level Unsupervised Adversarial Dropout: In order to further improve the potential ambiguous predictions caused by the large intraclass variances of RSIs, we apply unsupervised adversarial dropout to reconfigure the feature level of the network to dynamically generate an FUAD mask. However, at the feature level, our network is fully convolutional and RSIs show strong spatial correlation [38], so the feature map activation also has a strong correlation, and in this case, the DUAD is invalid.

To address the strong spatial correlation of DUAD, we design an FUAD for RSIs, which adversarially drops the activation value of the whole feature maps, rather than the individual activation value. In the case of the random dropout mask β^m, we define an FUAD mask β^{fuad} as

$$\begin{aligned}\beta^{\text{fuad}} &= \underset{||\beta - \beta^m||_2 \le \delta F}{\arg\max}\ E\big[G(x; \beta^m), G(x; \beta)\big] \\ &= \underset{||\beta - \beta^m||_2 \le \delta F}{\arg\max}\ E\big[G_2(\beta^m \odot G_1(x)), G_2(\beta \odot G_1(x))\big]\end{aligned} \tag{16}$$

where $E[\cdot, \cdot]$ indicates a divergence function, β^m is a sampled dropout mask, and F is the dropout layer dimension. The method to obtain the FUAD mask β^{fuad} is similar to DUAD.

C. Training Objective

Our method contains a classification loss, an MAD loss, a discriminative enhancement (DE) loss, a virtual adversarial perturbation (VAP) loss, and a self-motivated learning loss (SML). In the domain adaptation setting, we define the source domain $\mathcal{D}_s = \{X_s, Y_s\}$ and the target domain $\mathcal{D}_t = \{X_t\}$. We computer the cross-entropy on the source data

$$\mathcal{L}(\mathcal{D}_s) = -\mathbb{E}_{x_s, y_s \sim \mathcal{D}_s}\big[y_s \log G_2(x_s)\big] \tag{17}$$

where y_s is a one-hot encoded vector. For β^m and β^{fuad}, the feature extractor takes x_t and generates a feature vector, which is spread to a classifier to create two different predict distributions. At the feature level, in order to minimize the divergence between two different predict distributions, we employ β^{fuad} and the Kullback–Leibler (KL) divergence to calculate the FUAD loss

$$\begin{aligned}&\mathcal{L}_f(\mathcal{D}_t) \\ &= E_{x_t \sim \mathcal{D}_t}\big[D\big[G_2(\beta^m \odot G_1(x_t)), G_2(\beta^{\text{fuad}} \odot G_1(x_t))\big]\big] \\ &= E_{x_t \sim \mathcal{D}_t}\big[D_{\text{KL}}\big[G_2(\beta^m \odot G_1(x_t))||G_2(\beta^{\text{fuad}} \odot G_1(x_t))\big]\big].\end{aligned} \tag{18}$$

Similarly, for a target data point x_t, we apply decision level adversarial dropout β^{duad} in the classifier and the KL divergence to calculate the DUAD loss

$$\begin{aligned}&\mathcal{L}_d(\mathcal{D}_t) \\ &= E_{x_t \sim \mathcal{D}_t}\big[D\big[G_2(\beta^t \odot G_1(x_t)), G_2(\beta^{\text{duad}} \odot G_1(x_t))\big]\big] \\ &= E_{x_t \sim \mathcal{D}_t}\big[D_{KL}\big[G_2(\beta^t \odot G_1(x_t))||G_2(\beta^{\text{duad}} \odot G_1(x_t))\big]\big].\end{aligned} \tag{19}$$

The MAD loss consists of an FUAD loss and a DUAD loss

$$\mathcal{L}_{\text{MAD}}(\mathcal{D}_t) = \mathcal{L}_f(\mathcal{D}_t) + \mathcal{L}_d(\mathcal{D}_t). \tag{20}$$

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How to improve the insufficient exploration of discriminative feature representations for both domains caused by the large intraclass variances of RSIs is the key to the current task. We introduce the DE loss to reduce domain shift. This loss strengthens the samples with weak predictive ability in the target domain so that they have the stronger discriminative ability

$$\mathcal{L}_{\mathrm{DE}}(\mathcal{D}_t) = -E_{x_t \sim \mathcal{D}_t}[||G_2(x_t)||_*] \tag{21}$$

where $||\cdot||_*$ represents the nuclear norm.

VAP provides an orthogonal regularization to network parameter perturbation so that it generates perturbation more flexibly to enhance feature discriminative learning, leading to better generalization. The VAP loss is defined as

$$\mathcal{L}_{\mathrm{VAP}}(\mathcal{D}_t) = E_{x_t \sim \mathcal{D}_t}\left[\max_{||\lambda|| \le \eta} D_{\mathrm{KL}}[G_2(x_t)||G_2(x_t+\lambda)]\right] \tag{22}$$

where λ is the VAP on target images x_t.

In the target domain, how to make full use of the generated pseudo-labels by our network to boost the potential ambiguous discriminative learning of features is crucial. We introduce SML to generate the self-motivated $\bar{y}$ on target domain data

$$\bar{y} = \begin{cases} 1 & \text{if } c^* = \arg\max_c p_t,\ p_t > \eta \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

where η denotes the predefined threshold, we set $\eta = 0.95$. Then, our SML loss is defined as

$$\mathcal{L}_{\mathrm{SML}}(\mathcal{D}_t) = -E_{x_t, \bar{y} \sim \mathcal{D}_t}\left[\bar{y} \log G_2(x_t)\right]. \tag{24}$$

The total loss function is defined as

$$\mathcal{L}(\mathcal{D}_s, \mathcal{D}_t) = \mathcal{L}(\mathcal{D}_s) + \lambda_1 \mathcal{L}_{\mathrm{MAD}}(\mathcal{D}_t) + \lambda_2 \mathcal{L}_{\mathrm{DE}}(\mathcal{D}_t) + \lambda_3 \mathcal{L}_{\mathrm{VAP}}(\mathcal{D}_t) + \lambda_4 \mathcal{L}_{\mathrm{SML}}(\mathcal{D}_t) \tag{25}$$

where the hyper-parameters λ_1, λ_2, λ_3, and λ_4 are the weighting factor of the $\mathcal{L}_{\mathrm{MAD}}(\mathcal{D}_t)$, $\mathcal{L}_{\mathrm{DE}}(\mathcal{D}_t)$, $\mathcal{L}_{\mathrm{VAP}}(\mathcal{D}_t)$, and $\mathcal{L}_{\mathrm{SML}}(\mathcal{D}_t)$, respectively.

IV. Experiments

A. Datasets and Experimental Setup

We evaluate our method and many state-of-the-art domain adaptation algorithms on six adaptation scenarios, e.g., Merced → RSSCN7, AID → RSSCN7, NWPU → RSSCN7, RSSCN7 → Merced, RSSCN7 → AID, and RSSCN7 → NWPU. The RSSCN7 [39] dataset contains 2800 RSIs of seven scene categories. We choose RSSCN7 as the target domain or source domain. The Merced [40] dataset contains 2100 RSIs of 21 scene categories. The AID [41] dataset contains 10 000 RSIs of 21 scene categories. The NWPU [25] dataset consists of 31 500 RSIs of 45 scene classes. In RSSCN7, Merced, AID, and NWPU datasets have five public scene categories, namely, "Dense residential," "Forest," "Parking lot," "River" and "Agricultural." Therefore, we select these five categories from these datasets for domain adaptation scene classification. For evaluation, we use overall accuracy (OA) and kappa coefficient as evaluation metrics to assess the performance of the proposed method and several advanced algorithms.

B. Implementation Details

We use the PyTorch framework [42] for our implementation. All experiments are carried out on an NVIDIA 1080Ti GPU with 11 GB memory. We use a Stochastic Gradient Descent optimizer with a learning rate of 0.01, momentum 0.9, and weight decay 5×10^{-4}. For the backbone network, we use ResNet-50 [43] with a pretrain model to initialize the backbone network as the basic model. To maximize capturing feature discriminative representations, the CAFR is applied on the first conv layer's feature output and the last avg-pool's feature input for each domain. Specifically, for the target domain, we apply the decision level adversarial dropout after the second fully connected layer, and feature level adversarial dropout after the second CAFR module. For weighting factors of $\mathcal{L}_{\mathrm{MAD}}(\mathcal{D}_t)$, $\mathcal{L}_{\mathrm{DE}}(\mathcal{D}_t)$, $\mathcal{L}_{\mathrm{VAP}}(\mathcal{D}_t)$, and $\mathcal{L}_{\mathrm{SM}}(\mathcal{D}_t)$, we fix $\lambda_1 = 2$, $\lambda_2 = 0.025$, $\lambda_3 = 0.4$, and $\lambda_4 = 0.01$ in all our experiments.

C. Experimental Results and Comparisons

We compare the proposed method with other state-of-the-art domain adaptation approaches: Source-Only, ResNet-50 [43], ADR [44], DAN [45], JAN [23], MCD [46], iCAN [47], and DTA [48]. Table I shows all the experimental results when RSSCN7 is used as the target domain, and Table II shows all the experimental results when RSSCN7 is used as the source domain, where the best result is highlighted in each row in bold.

1) Merced → RSSCN7: Table I gives the result of our method and other state-of-the-art domain adaptation methods on the Merced → RSSCN7. We observe that our strategy brings about a noticeable improvement of 22.54% and 28.26% in terms of OA and Kappa compared to the Source-Only classification method, respectively, which obviously shows the advantage of our method in training a better generalization classifier under domain adaptation settings, while further shows the performance of scene classification is very different under the domain adaptation and fully supervised settings. Besides, our method also outperforms the ADR by 12.60% and 14.36%, DAN by 1.68% and 2.08%, JAN by 18.34% and 22.89%, DTA by 7.20% and 8.98%, iCAN by 8.19% and 18.89%, and MCD by 5.47% and 6.58% in terms of OA and Kappa, respectively. The reasons for this significant enhancement are as follows: 1) the CAFR module utilizes a self-gating mechanism to spontaneously recalibrate global and local features discriminative learning, such that the regions with discriminative global and local context information features can be discovered for source and target domains; 2) the MAD module strengthens the sparsity of our network to drop low-quality features in the target domain; and 3) the MAD loss and SML are introduced to further reduce ambiguities in the refinement.

2) AID → RSSCN7: Table I provides the comparative results on AID → RSSCN7. Our final model yields 90.67% and 88.31% in terms of OA and Kappa, which significantly improves the nonadaptive classification result by 8.07% and 10.06%. Besides, our method outperforms the current state-of-the-art method [44] by 7.73% and 9.65%, [45] by 0.01% and 0.03%, [23] by 4.07% and 5.06%, [48] by 1.72% and

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TABLE I

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF KAPPA AND OA VALUES ON THE MERCED → RSSCN7, AID → RSSCN7, AND NWPU → RSSCN7 DATASET

Data set	Class	Source-Only	ResNet-50	ADR	DAN	JAN	MCD	iCAN	DTA	Our method
Merced→RSSCN7	Dense residential	48.75	73.25	34.21	45.34	41.50	76.25	48.50	55.50	**95.25**
	Forest	76.00	97.25	91.43	98.22	80.00	95.75	95.50	96.50	**98.50**
	Parking lot	26.50	31.00	51.53	**86.54**	48.50	50.75	43.25	47.50	45.75
	River	81.75	89.20	94.04	91.06	84.25	86.84	70.75	**94.34**	89.46
	Agricultural	79.75	**98.75**	91.05	95.87	79.50	88.50	91.25	96.00	96.50
	Kappa (%)	53.07	72.28	66.97	79.25	58.44	74.75	62.44	72.35	**81.33**
	OA (%)	62.55	77.89	72.45	83.41	66.75	79.62	76.90	77.89	**85.09**
AID→RSSCN7	Dense residential	96.00	**98.25**	78.50	95.61	92.00	94.50	96.75	94.25	95.75
	Forest	90.00	**99.0**	93.75	99.60	80.75	87.50	92.75	94.0	98.25
	Parking lot	54.00	53.25	62.00	**98.72**	79.00	65.75	64.00	62.75	65.75
	River	80.98	91.26	87.69	74.88	87.75	82.37	**95.75**	95.00	95.12
	Agricultural	92.00	98.50	92.75	88.11	93.50	96.25	97.00	**98.75**	98.50
	Kappa (%)	78.25	85.04	78.66	88.28	83.25	81.62	86.56	86.17	**88.31**
	OA (%)	82.60	88.05	82.94	90.66	86.60	85.27	90.10	88.95	**90.67**
NWPU→RSSCN7	Dense residential	82.25	97.50	82.00	91.57	86.18	90.50	89.50	93.00	**98.00**
	Forest	91.50	96.50	89.75	96.71	87.32	92.75	90.25	96.50	**98.50**
	Parking lot	**95.50**	79.25	91.50	90.29	89.45	87.50	89.25	78.00	79.00
	River	69.67	87.15	89.45	93.43	82.05	87.63	92.50	90.49	**94.34**
	Agricultural	90.00	92.00	81.25	78.43	91.03	86.50	90.75	**93.75**	93.00
	Kappa (%)	82.33	88.12	83.48	87.61	84.04	86.24	88.06	88.93	**90.70**
	OA (%)	85.78	90.48	86.79	90.09	87.21	88.98	90.45	90.35	**92.57**

TABLE II

PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF KAPPA AND OA VALUES ON THE RSSCN7 → MERCED, RSSCN7 → AID, AND RSSCN7 → NWPU DATASET

Data set	Class	Source-Only	ResNet-50	ADR	DAN	JAN	MCD	iCAN	DTA	Our method
RSSCN7→Merced	Dense residential	91.46	78.24	69.16	45.34	49.64	71.20	91.06	74.25	**94.31**
	Forest	92.08	92.67	88.23	**98.22**	98.03	92.27	93.24	97.22	98.11
	Parking lot	87.11	61.08	**97.52**	86.54	83.24	60.24	76.85	62.34	72.61
	River	67.80	89.83	88.37	91.06	89.13	78.75	84.93	**91.53**	83.05
	Agricultural	66.24	94.03	93.10	95.87	**96.22**	92.07	94.88	94.09	94.30
	Kappa (%)	77.56	79.94	84.04	79.25	79.04	73.62	85.25	78.85	**85.41**
	OA (%)	80.94	83.17	87.28	83.41	83.25	78.91	88.19	83.88	**88.47**
RSSCN7→AID	Dense residential	**99.76**	88.29	93.90	95.61	96.34	92.20	94.15	87.80	96.51
	Forest	89.20	98.80	98.80	**99.60**	98.00	92.80	92.80	99.20	96.20
	Parking lot	76.67	98.46	94.10	98.72	98.21	96.67	90.51	**98.72**	93.15
	River	73.70	83.01	84.06	74.88	77.56	79.74	81.46	79.73	**93.99**
	Agricultural	80.81	93.51	82.97	88.11	92.16	88.65	91.89	**93.78**	90.54
	Kappa (%)	79.80	89.93	87.79	88.28	89.85	87.28	87.28	89.16	**92.67**
	OA (%)	84.03	92.41	90.77	91.38	92.45	90.01	90.16	91.85	**94.08**
RSSCN7→NWPU	Dense residential	93.14	97.71	93.14	91.57	90.86	**98.57**	90.57	96.86	97.43
	Forest	91.00	89.86	96.71	96.71	**97.14**	87.29	97.00	90.43	89.25
	Parking lot	94.29	98.86	94.29	90.29	90.71	96.43	95.85	98.43	**99.51**
	River	70.76	88.17	92.24	93.43	93.29	84.26	88.00	77.69	**96.26**
	Agricultural	85.14	94.43	81.86	78.43	80.43	85.57	81.57	**97.00**	92.71
	Kappa (%)	83.81	92.32	89.59	87.61	88.11	88.07	88.25	90.26	**93.80**
	OA (%)	86.87	93.81	91.65	90.09	90.49	90.42	90.60	92.08	**95.03**

2.14%, [47] by 0.57% and 1.75%, and [46] by 5.40% and 6.69% in terms of OA and Kappa, respectively. These results suggest that the effective combination of our CAFR and MAD modules can substantially promote classification accuracy. This also implies that our method has a good generalization ability in different domain adaptation scenarios, and further shows that our method can improve the ability of the discriminative feature representations when faced with the large intraclass variances of RSIs.

3) NWPU → RSSCN7: We report the quantitative evaluation on NWPU → RSSCN7 of our method and state-of-the-art methods in Table I. We can observe that our method brings a large improvement of 5.78% and 7.22% to ADR, 2.48% and 3.09% to DAN, 5.36% and 6.66% to JAN, 3.59% and 4.46% to MCD, 2.12% and 2.64% to iCAN, and 2.22% and 1.77% to DTA in terms of OA and Kappa, respectively. Besides, our method outperforms the Source-Only by 6.79% and 8.37% in terms of OA and Kappa. For "Dense residential," "Forest," and "River," we can achieve superior classification accuracy compared with the most advanced methods. Behind these good results, we can attribute them to the CAFR module spontaneously mining the global and local discriminative context information of features with a self-gating mechanism, while the MAD module further strengthens the robustness of features by dynamically adjusting the sparsity of the feature distributions of feature level and decision level.

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TABLE III
RESULTS ON THE MERCED → RSSCN7, AID → RSSCN7, AND NWPU → RSSCN7 DATASET FOR EACH KEY COMPONENT

Merced→RSSCN7							
Architecture	Context-aware feature refinement module		√	√	√	√	√
Loss Function	Discriminative enhancement loss	√		√	√	√	√
	Virtual adversarial perturbation loss	√	√		√	√	√
	self-motivated learning loss	√	√	√		√	√
Multi-level adversarial dropout module		√	√	√	√		√
OA (%)		77.89	50.53	83.51	50.53	69.63	85.09
Kappa (%)		72.28	38.22	79.39	38.22	62.04	81.33
AID→RSSCN7							
Architecture	Context-aware feature refinement module		√	√	√	√	√
Loss Function	Discriminative enhancement loss	√		√	√	√	√
	Virtual adversarial perturbation loss	√	√		√	√	√
	self-motivated learning loss	√	√	√		√	√
Multi-level adversarial dropout module		√	√	√	√		√
OA (%)		88.05	72.45	89.69	89.59	88.89	90.67
Kappa (%)		85.04	65.56	87.12	86.99	86.11	88.31
NWPU→RSSCN7							
Architecture	Context-aware feature refinement module		√	√	√	√	√
Loss Function	Discriminative enhancement loss	√		√	√	√	√
	Virtual adversarial perturbation loss	√	√		√	√	√
	self-motivated learning loss	√	√	√		√	√
Multi-level adversarial dropout module		√	√	√	√		√
OA (%)		90.48	78.88	91.10	91.96	91.15	92.56
Kappa (%)		88.12	73.59	88.87	89.94	88.94	90.70

4) RSSCN7 → Merced: We choose RSSCN7 as the source domain. Table II presents the quantitative evaluation on RSSCN7 → Merced of the experimental results. We notice that our method obtains a significant improvement of 7.53% and 7.85% in terms of OA and Kappa compared to the Source-Only classification method, respectively, which shows our learning method can train a better generalization classifier under domain adaptation settings. Besides, our method outperforms the ADR by 1.19% and 1.37%, DAN by 5.06% and 6.16%, JAN by 5.22% and 6.37%, DTA by 4.59% and 6.56%, iCAN by 0.28% and 0.13%, and MCD by 9.56% and 11.79% in terms of OA and Kappa, respectively. This is mainly because our method exploits the truly useful discriminative feature representation information to better adapt our classifier.

5) RSSCN7 → AID: The experimental results of RSSCN7 → AID are shown in Table II. Our train model outputs 94.09% and 92.67% in terms of OA and Kappa, which effectively improves the nonadaptive classification result by 10.05% and 12.87%. Beyond that, our method is superior to the current state-of-the-art method [44] by 1.67% and 2.74%, [45] by 2.7% and 4.39%, [23] by 1.63% and 2.82%, [48] by 2.23% and 3.51%, [47] by 3.92% and 5.39%, and [46] by 4.07% and 5.39% in terms of OA and Kappa, respectively. These encouraging results verify that our method can effectively derive a robust classifier.

6) RSSCN7 → NWPU: In Table II, we can see the experimental results on RSSCN7 → NWPU. Our method is superior to other comparative experimental results. For the "Parking lot" class and "River" class, compared with the most advanced classification methods, we can obtain higher classification accuracy. These improvements mean the effectiveness of our method. This is mainly due to the fact that the strong combination of the MAD module and CAFR module activates more potential discriminative knowledge in the network and improves the generalization capability of the model.

In general, our method has not only achieved great success on Merced → RSSCN7, AID → RSSCN7, and NWPU → RSSCN7 but also good generalization ability on RSSCN7 → Merced, RSSCN7 → AID, and RSSCN7 → NWPU. Therefore, the domain discrepancy between the source and target domain is greatly improved. However, in Table I, our method fails in the "Parking lot" class. Maybe it is because our method mistakenly treats the car-related area as the "Parking lot" class and leads to the misalignment of source and target domain features.

D. Ablation Study

Although our method shows significant improvements on multiple domain adaptation datasets, we want to examine the role of proposed each component in practice. A chain of ablation experiments is carried out and the results are shown in Table III. To disclose the contribution of each component in our method, we conduct ablation experiments to comprehensively evaluate the contributions of each component on the Merced → RSSCN7, AID → RSSCN7, and NWPU → RSSCN7 datasets.

To assess whether each module of our method is important to performance, we study the influence of each component on our experimental method. We can see that the proposed

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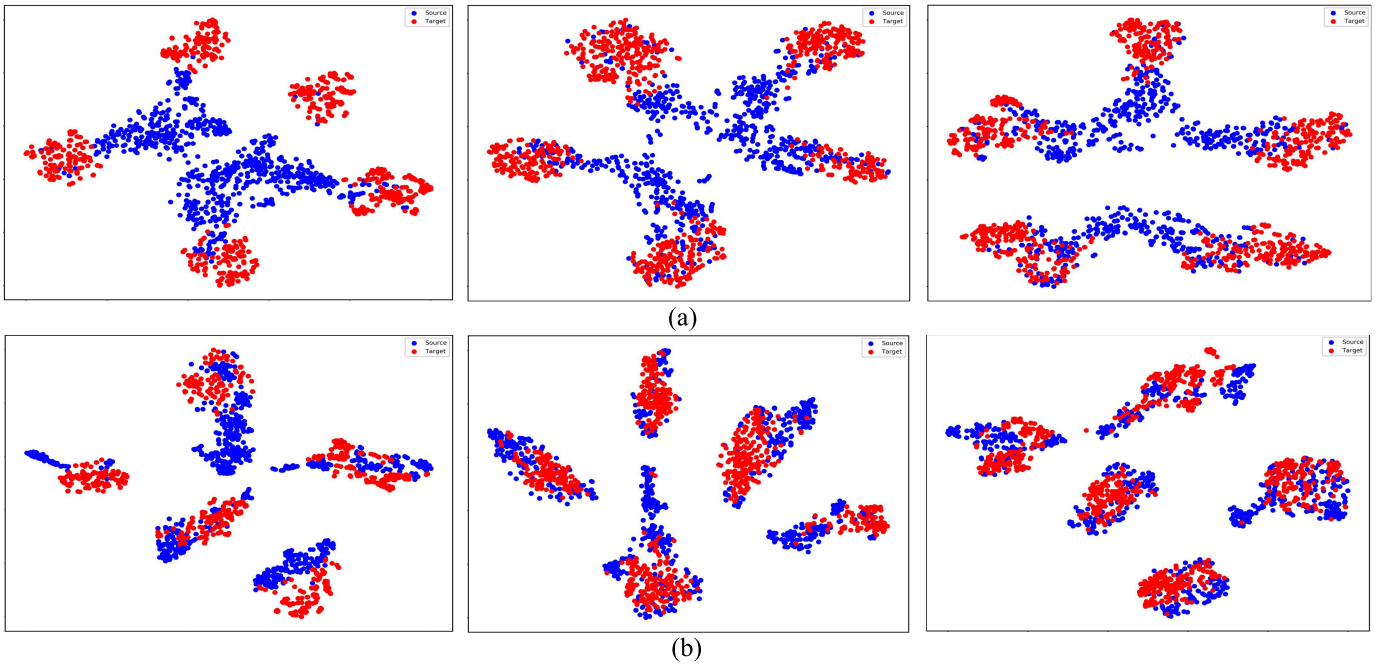


Fig. 4. t-SNE visualization of Merced → RSSCN7, AID → RSSCN7, and NWPU → RSSCN7 classification dataset using our network, before and after adaptation with our method. t-SNE hyper-parameters are consistent in both visualizations. (a) Source-only. (b) Our method.

components have different contributions in Table III. Results in Table III show that the CAFR module largely improves the classification accuracy. It is mainly because the CAFR effectively activates the global and local features of high quality by a self-gating mechanism to "automatically" emphasize valuable feature information and suppress less useful ones according to capture the long-range dependency. As shown in Table III, we can observe the following. First, the designed MAD module on three domain adaptation datasets can offer 15.46%, 1.75%, and 1.41% in terms of OA gains and the Kappa gains improve by 19.29%, 2.20%, and 1.76%, respectively. A possible explanation is that the MAD module increases the sparsity of our network by adaptively reconfiguring the feature level and decision level of the network in the target domain, which enables more valuable discriminative features to be activated. Second, our DE loss on three domain adaptation datasets brings a marked improvement of 34.56%, 18.22%, and 13.68% in terms of OA and 43.11%, 22.75%, and 17.11% in terms of Kappa, which explicitly proves the superiority of our DE loss for training a better discriminative classifier under domain adaptation settings. Third, the VAP loss on three domain adaptation datasets improves 1.51%, 0.98%, and 1.46% in terms of OA and 1.94%, 1.19%, and 1.83% in terms of Kappa, which illustrates that our VAP loss provides an orthogonal regularization to network parameter perturbation so that it generates perturbation more flexibly to enhance feature discriminative learning. Fourth, the SML loss on three domain adaptation datasets achieves 4.80%, 1.08%, 1.08%, 0.60% improvement in terms of OA and 5.97%, 1.32%, and 0.76% improvement in terms of Kappa, which shows that SML guides domain adaptation to enhance the network's generalization capability.

E. Feature Qualitative Analysis

To get a deeper understanding of our method, we exhibit the feature distributions before and after adaptation of Merced → RSSCN7, AID → RSSCN7, and NWPU → RSSCN7 in the 2-D space in Fig. 4. We exploit the t-distributed stochastic neighbor embedding (t-SNE) [49] to visualize embedded feature distributions between the source and target domains. Fig. 4(a) shows the feature distribution while only operating on the source domain data. From Fig. 4(a), we can see that there is a large domain discrepancy between the source and target domains. In Fig. 4(b), we find that our learning method mitigates the data gap between the source and target domains, and the prediction model produced by our method has a stronger generalization ability. In addition, our method shows a better ability to solve the large intra-class variances of RSIs, which indicates the proposed components can promote the network to learn highly discriminable representations.

V. Conclusion

This article proposes a novel end-to-end domain adaptation scene classification network for RSIs, named DFENet, which consists of a CAFR module and an MAD module. The CAFR module is constructed to spontaneously recalibrate global and local features by modeling feature responses between the channel and spatial for each domain. Then, the MAD module is further developed to increase the generalization performance by spontaneously reconfiguring the sparsity of our network structure so that more high-quality discriminative features are activated. The collaboration of CAFR and MAD facilitates the robustness of the scene classification model. In our experiments on Merced → RSSCN7, AID → RSSCN7,

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NWPU → RSSCN7, RSSCN7 → Merced, RSSCN7 → AID, and RSSCN7 → NWPU datasets, we validate that the proposed method significantly outperforms state-of-the-art methods.

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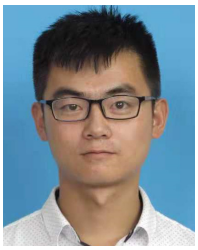
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